



Contents lists available at ScienceDirect

Resources, Conservation & Recycling

journal homepage: www.sciencedirect.com/journal/resources-conservation-and-recycling

Editorial

Establishing resilient and sustainable supply chain of critical materials for a low-carbon future

1. Introduction

The transition to clean energy and carbon neutrality hinges on reliable access to critical raw materials, such as lithium, cobalt, nickel, rare earth elements, etc., that are essential for technologies like lithium-ion batteries (LIBs), fuel cells, wind turbines, solar photovoltaics, and electric vehicles (EVs) (Sun et al., 2024; United Nations, 2025). Global demand for these minerals is rising rapidly as decarbonization accelerates. For example, the International Energy Agency projects that mineral requirements for clean energy technologies could almost triple by 2030 and reach about 40 million tonnes under a net-zero scenario by 2040 (IEA, 2021). The extraction and processing of these critical materials are frequently associated with severe sustainability burdens (Giljum et al., 2025). For example, lithium mining requires over 1000 times more freshwater than iron extraction per unit of material produced. This leads to large concerns about water depletion and habitat destruction from expanding lithium mining in arid regions (Sun et al., 2025a). Cobalt mining in the D.R. Congo has been linked to human rights issues and pollution (Hao and Sun, 2021). Nickel extraction often entails large-scale land disturbances, making it a major driver of tropical deforestation in Indonesia (Giljum et al., 2022). Therefore, the surge in critical material demand makes sustainable supply chains a paramount concern: ensuring sufficient supply of critical materials, while minimizing environmental and social impacts, is now recognized as an “essential task” for achieving climate targets. At the same time, policymakers are increasingly aware that a renewables-intensive energy system comes with new vulnerabilities; that is, issues of price volatility, geographic concentration, and geopolitical risks in mineral supply can threaten energy security just as seriously as fossil fuel disruptions (European Commission, 2024; Trost and Dunn, 2023). In short, securing critical material supply chains has become indispensable for clean energy transitions and carbon neutrality efforts worldwide.

Against this backdrop, the special issue “Establishing Resilient and Sustainable Supply Chain of Critical Materials for a Low-carbon Future” brings together recent research aimed at understanding and addressing these challenges. This special issue comprises 24 articles that illuminate strategies to sustainably meet booming material demand and fortify supply chains against disruptions. Notably, the contributions coalesce around five interrelated themes: (1) supply-demand dynamics projection, (2) clean production and circular economy strategies, (3) material criticality and supply chain resilience, (4) mapping material footprint, supply, and value chains, and (5) policy and governance. In the rest of this editorial, we briefly review each thematic area and highlight the key insights, novel methods, and conclusions offered by the contributing authors, before reflecting on the overarching lessons and future

directions.

1.1. Supply-demand dynamics projection

Several studies on this special issue underscore the magnitude of impending demand growth and the importance of recycling, reuse, and alternative sourcing to meet this demand. Given the rapid expansion of solar photovoltaics and other uses, Cattaneo et al. (2026) forecast global demand and supply of silver to 2030. Their analysis warns of a potential silver supply shortfall by 2030. They emphasize that mitigating this supply risk will require aggressive circular economy measures and material efficiency: reducing silver usage in all applications, identifying suitable substitute materials, and expanding secondary production are essential strategies to avoid bottlenecks in the solar energy transition.

Similar concerns are echoed for other critical materials. Simas et al. (2025) present a global value-chain assessment of material needs for next-generation LIB technologies. By linking material flow analysis with multi-regional input-output models, they evaluate how shifting battery chemistries (e.g., high-nickel, cobalt-free, solid-state designs) could alter the demand for lithium, cobalt, nickel, graphite, and other inputs across international supply chains. This approach reveals potential hotspots in the supply network and highlights the importance of upstream planning: anticipating the material requirements of new battery chemistries can inform timely investments in mining, recycling, or substitution to prevent future shortages. In a related vein, Zhen et al. (2025) focus on China’s ambitious hydrogen energy roadmap and caution that the supply of platinum group metals for electrolyzers and fuel cells could constrain the scale-up of hydrogen production. They call for proactive measures, including recycling and material substitution, to ensure that metal supply risks do not become a stumbling block in China’s and the world’s low-carbon hydrogen transition. Zong et al. (2025) combine criticality assessment with material flow analysis for the raw materials used in China’s burgeoning power LIB industry, examining scenarios “towards sustainable supply and recycling”. They identify which battery materials are most critical regarding supply risk and then evaluate how improved recycling rates and material substitution in batteries could alleviate these risks. This integrated analysis helps pinpoint where policy interventions would yield the greatest resilience benefits for China’s EV battery supply chain.

Another group of papers under this theme centers on how innovations in product design, material substitution, and efficiency improvements can reduce the strain on critical material supply. One clear avenue for improvement is enhancing the material efficiency of technologies. Hoff et al. (2025) provide a striking example of the aviation industry’s use of titanium. They analyze several circular strategies for

<https://doi.org/10.1016/j.resconrec.2025.108657>

Received 2 October 2025; Accepted 21 October 2025

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reducing primary titanium demand, including recycling retired aircraft, extending aircraft service lifetimes, and an “enhanced” lifetime extension combined with engine retrofits. Their findings show that recycling alone would have only a marginal effect, while pure lifetime extension offers the greatest benefit. These results underscore the complexity of circular strategies: not all approaches deliver equal gains, and the most impactful strategy for critical metals in high-tech sectors may be to extend the usable life of products. In the realm of EVs, [Sullivan et al. \(2025\)](#) introduce a novel bottom-up modeling approach to examine how automakers’ design and technology choices affect lithium demand and price elasticity. Their model integrates detailed insights from automaker interviews about current and emerging battery technologies into an industry-wide demand forecast for lithium. A key insight is that in the short run, greater optionality in EV technologies leads to higher lithium price elasticity. In the long run, as EV adoption saturates and technology choices consolidate, the model shows elasticity first rising but then declining. These findings suggest that maintaining technological diversity in the EV market acts as a buffer, increasing the adaptability of material demand to supply conditions.

In addition to the material substitution example in the silver PV study of [Cattaneo et al. \(2026\)](#), [Sun et al. \(2025b\)](#) echo the call for substitution in their study of critical minerals for China’s vehicle electrification, explicitly proposing eco-design strategies to reduce dependency on the highest-risk materials. They couple these design suggestions with risk analysis, meaning the choices are informed by which materials pose the greatest threat to supply or sustainability. This kind of integrated eco-design approach exemplifies how engineers and product developers can contribute to supply chain resilience: by reimagining products to use earth-abundant materials, to be more modular, or to be easier to recycle, thereby cutting down critical raw material requirements.

Combining a national energy system model with life-cycle inventory data on material inputs, [Hahn Menacho et al. \(2025\)](#) provide a holistic view of how pursuing climate goals might shift burdens onto material supply chains, using Switzerland as a case study. Their scenario analysis shows that even as operational greenhouse gas emissions plummet, the demand for many critical materials will rise substantially in a net-zero energy system. They highlight that raw material requirements must be integrated into climate and energy planning and potential environmental trade-offs: the lifecycle emissions and impacts associated with mining lithium or nickel could erode the benefits of decarbonization if not managed correctly. [Parpan et al. \(2026\)](#) develop a mine-level dynamic modeling approach to assess whether future primary copper supply can meet the socioeconomic trajectories embedded in the Shared Socioeconomic Pathways, combining commercial mine-by-mine data with the DyMEMDS demand model. Their results show that several high-growth socioeconomic pathways are unlikely to be compatible with expected copper availability, underscoring a fundamental inconsistency between decarbonization ambitions and upstream resource limits. The study highlights the need for integrating mining constraints, circularity, and material efficiency into decarbonization modeling to avoid unrealistic pathways.

1.2. Clean production and circular economy strategies

Two studies have demonstrated the potential of cleaner production pathways for lithium based on emerging technologies. [Mousavinezhad et al. \(2025\)](#) evaluate a novel method for sustainable lithium production from sedimentary clay deposits, assessing its carbon footprint based on the life cycle assessment (LCA) and integration with EV supply chains. This study points to diversifying lithium sources beyond conventional brines and hard-rock mines: sedimentary deposits, if extracted with low-carbon processes, could both enlarge supply and reduce overall environmental impacts. [Nishikawa et al. \(2026\)](#) conduct a cradle-to-gate life cycle assessment of novel North American lithium production pathways, which is using direct lithium extraction from geothermal

brines, rock-based extraction with electrochemical refining, and hydrometallurgical recycling to produce $\text{LiOH} \cdot \text{H}_2\text{O}$. They show that low-carbon electricity is the main determinant of emissions, and that recycling can outperform mining-based routes under certain allocation approaches. Expanding the boundary to battery manufacturing reveals that upstream lithium innovations offer modest benefits compared to circular strategies.

Multiple studies investigate circular economy pathways for critical materials to reduce reliance on virgin raw materials. [Wang et al. \(2025\)](#) conduct a critical life-cycle assessment of current and potential rare earth element (REE) recycling routes from permanent magnets, which are widely used in wind turbines and EV motors. By comparing the environmental impacts of different end-of-life scenarios, they identify opportunities for substantial impact reductions if efficient magnet-to-magnet recycling loops can be established. Similarly, [Joshi et al. \(2025\)](#) demonstrate an innovative bioleaching approach to recover REEs from industrial wastes. Utilizing microorganisms to extract rare earths from waste streams offers a more sustainable alternative to conventional mining, potentially closing the loop for these critical elements. Their study highlights that biohydrometallurgical techniques can achieve meaningful REE recovery yields while avoiding the harsh chemicals and high emissions of traditional extraction, illustrating a promising circular strategy for critical materials. [Finley et al. \(2025\)](#) contribute another piece to this puzzle by optimizing aqua regia-based digestion methods for recovering trace rare earth elements from coal fly ash. Finley and colleagues test improved leaching techniques, including the use of aqua regia under microwave heating, to extract REEs from fly ash of different classes. Their results demonstrate that substantial fractions of certain REEs can be liberated from ash, depending on its composition, potentially turning a large-volume waste into a new unconventional source of critical elements.

A systematic perspective on the corporate adoption of circular strategies is provided by [Otterbach and Fröhling \(2025\)](#). They present a comprehensive systematic review of how raw material criticality assessments support corporate circular economy decisions. They find that while criticality assessments have become a popular tool for companies to identify which materials are “critical” to their operations, there is significant variation in approaches and a need for better integration of circular economy considerations. They highlight that next-generation criticality frameworks should incorporate life cycle and socio-environmental factors aligning with LCA and business sustainability strategies, so companies can use them to prioritize investments in recycling technologies, develop substitution R&D programs, or redesign products for improved resource efficiency.

Another important thread in this theme is the global trade and equity dimension of circular economy practices. [Zisopoulos et al. \(2025\)](#) offer a complementary perspective by examining inequities in the global trade of waste metals through a bio-inspired network analysis. They find that between 2000 and 2022, global trade in metal waste grew substantially in volume and value, yet the circulation efficiency of these flows remains relatively low and has even declined. Their analysis shows the network is robust but not “vital” in ecological terms, dominated by a few high-income countries that import high-value scrap from lower-income exporters. They call for international policy measures to tackle these circularity and trade inequities, including incorporating environmental and social safeguards across product lifecycles.

[Cristiu et al. \(2025\)](#) offer a novel insight into optimally designing recycling supply chains. They formulate an optimization model to locate and size recycling facilities under uncertainty. These technological innovations will be central to a sustainable materials future. By improving the efficiency and cost-effectiveness of material recovery from end-of-life products, industrial wastes, or low-grade ores, design innovations in processing can reduce the need for new mining and alleviate supply bottlenecks.

1.3. Material criticality and supply chain resilience

Another group of papers delves into supply chain resilience. At the conceptual level, [Wietschel et al. \(2025\)](#) propose an integrated framework for raw material criticality and supply chain resilience assessments. Criticality assessments and resilience assessments have traditionally been separate analytics. They argue that combining these perspectives can yield a more comprehensive evaluation of where a supply chain is truly vulnerable and why. They outline a framework that links factors like supply concentration, disruption probability, and recovery capacity, thereby bridging the concepts of critical raw materials and resilient supply chains. This framework helps decision-makers identify which materials are at high risk of shortage, how susceptible the entire supply chain is to shocks, and what adaptive capacities exist or need improvement at each stage.

Other studies develop quantitative models tailored to specific supply chains. [Liu et al. \(2025\)](#) introduce a novel assessment model for China's cobalt supply chain resilience. They structure the complex supply chain issues into Drivers–Pressures–State–Impact–Response (DPSIR) categories and then apply machine learning techniques to analyze large datasets (e.g., historical production, trade, price, and policy data) and to identify patterns or early-warning signals of disruption. This data-driven approach yields insights into key risk factors and suggests targeted interventions for each stage. This DPSIR+ML framework represents a new decision-support tool for policymakers, enabling more predictive and evidence-based risk management for critical materials.

The EV supply chain emerges as a recurring case study in resilience analyses. [Lou et al. \(2025\)](#) conduct a detailed resilience assessment of the EV supply chain under upstream mineral shortages. They simulate scenarios where production at key upstream nodes is curtailed and then quantify the impacts on downstream battery manufacturing and EV production. Their results likely quantify metrics like drop in output, time to recovery, and “resilience loss” under different disruption scenarios. A major finding is that bottlenecks in upstream mineral supply can quickly propagate downstream, given the limited short-term substitutability of critical minerals. They also examine strategies to improve resilience, illustrating how a combination of moderate strategic inventories and diversified sourcing could substantially reduce the production losses in an extreme disruption scenario. The importance of risk assessment tools in guiding mitigation strategies is further underlined by the above-mentioned review of [Otterbach and Fröhling \(2025\)](#). In addition to the circular economy side, from a risk management perspective, their review highlights that companies are increasingly using multi-criteria criticality assessments to pre-emptively identify materials that could jeopardize production continuity or corporate sustainability goals. They conclude that when criticality assessment is aligned with corporate strategy, it can serve as an early warning system and a decision-support tool: firms can prioritize which materials to focus their risk mitigation efforts on.

1.4. Mapping material footprint, supply, and value chains

A typical theme is the characterization of the global supply chain by mapping the suppliers, consumers, and trade pathways. [Su et al. \(2025\)](#) map the evolution of the global nickel cycle and trade networks, using trade-linked material flow analysis and network metrics to uncover shifting patterns over time. Their work reveals an increasing interconnection of nickel supply chains worldwide, with some countries emerging as dominant recycling hubs and others as persistent net importers/exporters. Such insights into global material circulation can inform coordinated strategies to improve circularity, such as fostering international recycling partnerships or standardizing material quality for recycled nickel.

[Liu and Domenech Aparisi \(2025\)](#) investigate the global sustainability implications of China's rapidly growing LIB sector through a value chain analysis that integrates both physical and monetary flows.

By incorporating the concept of value-retaining processes, the study moves beyond traditional material flow assessments to capture where economic value is concentrated within the supply chain. The results show that lithium chemical refining and final battery-powered product manufacturing are the most significant stages of value creation, underscoring their importance for sustainability strategies. This dual perspective reveals that enhancing circular practices in these high-value segments could deliver outsized benefits, offering a framework for aligning industrial growth with sustainability goals.

[Zhang et al. \(2025\)](#) provide another perspective on material footprint from the technology level. They conduct a global life-cycle material footprint analysis of geothermal power generation technologies, comparing flash steam, dry steam, and binary cycle systems. By examining material requirements across all life-cycle stages from plant construction to operation and decommissioning, the study identifies distinct material intensity patterns for each technology and highlights the stages where material demand is most significant. The findings reveal that material footprints are not uniform: each technology exhibits unique hotspots, pointing to opportunities for optimization in design and material use. These insights show how targeted improvements in resource-intensive stages could enhance geothermal power's material efficiency and sustainability.

1.5. Policy and governance

[Ray et al. \(2025\)](#) zoom out to the critical mineral supply chains' broader policy and governance context. They explore how public financial institutions can support more sustainable and inclusive mineral supply chains through “patient” capital. Based on a systematic review and stakeholder interviews, they find that public development finance institutions and export credit agencies influence critical mineral supply chains via three main pathways. They identify notable gaps in the current practices of these financial actors. For example, policy support is frequently directed toward low-income countries in general, missing many mineral-rich countries that are middle-income. To close these gaps, the authors recommend targeted actions such as developing transparent sustainability metrics for green finance in mining and extending development finance support to critical mineral governance in resource-rich countries.

2. Conclusion and outlook

The articles in this special issue significantly advance our understanding of how to build resilient and sustainable supply chains for critical materials in a low-carbon future. First, there is a clear recognition that simply scaling up mining to meet burgeoning demand is neither environmentally nor geopolitically tenable; instead, a combination of demand-side management and circular economy strategies will be necessary to close the looming gap between critical material supply and demand. Besides, while recycling is crucial, it is often insufficient alone. Deeper changes in product design, like extending lifetimes, reducing material content, and changes to consumer usage patterns, will amplify the impact of recycling and are vital to achieve meaningful reductions in primary material needs.

Second, the importance of anticipatory risk assessment and resilience planning is highlighted throughout. Whether at the firm level, using criticality assessments to guide strategy, the supply chain level, modeling propagation of disruptions and identifying weak links, or the policy level, strengthening institutional capacity via development banks, proactive risk management is shown to pay dividends. By identifying potential chokepoints and failure modes in advance, stakeholders can implement mitigation measures before crises hit, for example, qualifying multiple sources for a material, holding strategic reserves, or investing in substitute technologies as insurance. One lesson is that resilience is not automatic: it must be deliberately engineered through strategic choices and investments.

Third, a recurring theme is the need for innovation and cross-sector collaboration. The special issue papers underscore that ensuring sustainable material supply is as much a technical challenge as an economic and policy challenge. On one hand, breakthroughs in recycling technology, materials science for substitutes, and manufacturing design will open new avenues to reduce pressure on critical minerals. On the other hand, aligning financial incentives and international policies, such as development finance, with sustainability goals or trade policies addressing inequities in waste flows will be essential to scale up those technical solutions globally. Collaboration between traditionally siloed domains, including energy planners working with mineral economists, manufacturers with recyclers, governments with industry and finance institutions, is a prerequisite for success.

In reflecting on the collective contribution of these studies, several key priorities for future research and action stand out:

- **Data and monitoring:** Improving the quality and transparency of data on critical material flows, stocks, and supply chain conditions. Several authors noted data limitations in supply chain traceability or recycling rates. Developing real-time monitoring and early-warning systems can greatly enhance preparedness.
- **Holistic assessment tools:** Integrative assessment tools that combine technical, economic, and environmental metrics of critical material supply dynamics are needed. Future research could create dynamic models that simulate market and supply chain conditions under various disruption scenarios, including feedback loops between material availability and technology adoption.
- **Material substitution and efficiency:** Another important area is accelerating R&D into alternative materials and product designs that achieve the same functionality with less material. The more substitution options exist, the more flexibility and elasticity the system gains. Policies like R&D funding, innovation prizes, or procurement standards could spur progress here.
- **Circular infrastructure scaling:** Many circular strategies have been proven in concept but need to be scaled up. This entails investment in collection systems, reprocessing facilities, and market development for secondary materials. Governments can aid by treating recycling and urban mining as critical infrastructure, simplifying permitting, providing incentives, and facilitating public-private partnerships.
- **International coordination:** International cooperation is crucial since critical mineral supply chains are global. This includes sharing best practices on sustainability standards, cooperating on stockpiling and emergency response, and ensuring that resource-rich developing countries can climb the value chain rather than just exporting raw materials. Platforms like the EU's Critical Raw Materials Act, the U.S. and allies' Minerals Security Partnership, or multilateral forums could incorporate the insights from these studies to pursue joint strategies.
- **Sustainable development and inclusion:** Pursuing critical minerals should not repeat the mistakes of past resource booms. Future efforts must integrate strong environmental and social governance, including protecting local communities, ensuring fair benefit-sharing, and minimizing ecological damage. This will require technical fixes and institutional innovations in how projects are financed and regulated.

The body of research in this special issue paves the way for a more resilient and sustainable critical materials supply chain by emphasizing these priorities. While the challenge is undoubtedly immense, the solutions are within reach if we mobilize knowledge from all quarters. Each article contributes a piece of the puzzle, forming a clearer picture of the path forward. As we push towards a low-carbon future, we hope this special issue will inform and inspire the next generation of research, policy, and practice that will turn resilient and sustainable supply chains from aspiration into reality. The stakes for climate mitigation and sustainable development could not be higher, but armed with the insights

from these works, we are better prepared to meet the challenge.

CRediT authorship contribution statement

Xin Sun: Conceptualization, Investigation, Methodology, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Christoph Helbig:** Writing – review & editing. **Han Hao:** Writing – review & editing. **Takuma Watari:** Writing – review & editing. **Karan Bhuwanka:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The guest editors extend their sincere appreciation to all authors and reviewers for their valuable contributions to this special issue. We also express our gratitude to Prof. Ming Xu, Editor-in-Chief of Resources, Conservation & Recycling, for his guidance in formulating the proposal and overseeing the organization of the VSI. Finally, we acknowledge with great thanks the editorial assistance of Prof. Zhi Cao in helping organize the VSI and preparing this concluding article.

Data availability

No data was used for the research described in the article.

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